A Three-Stage Quantitative Research Pipeline: Signal Discovery, Strategy Construction, and Portfolio Execution

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Project Overview

Many quantitative projects stop once a model looks good on historical data. This project asks a more practical question: **which signals still matter after they are turned into trades, combined into a portfolio, and charged realistic execution costs?**

The project studies daily ETF data from 2014 through 2025 across SPY, TLT, GLD, USO, UUP, EFA, EEM, and FXE. It follows a complete path: first test whether market structure exists, then convert selected signals into trading rules, and finally evaluate whether the surviving strategies improve a portfolio after turnover caps, transaction costs, slippage, and volatility targeting.

The full technical research report, including implementation details and extended results, is available here.

The evidence is strongest on risk management. Volatility and regime signals were useful, while the EFA–EEM amongst other pairs failed the research filter and should not be trusted as a mean-reversion trade under the current evidence. At the portfolio level, inverse-volatility weighting was the most robust selected allocator, while Markowitz variants needed regularization or hard caps to reduce noisy reallocations.

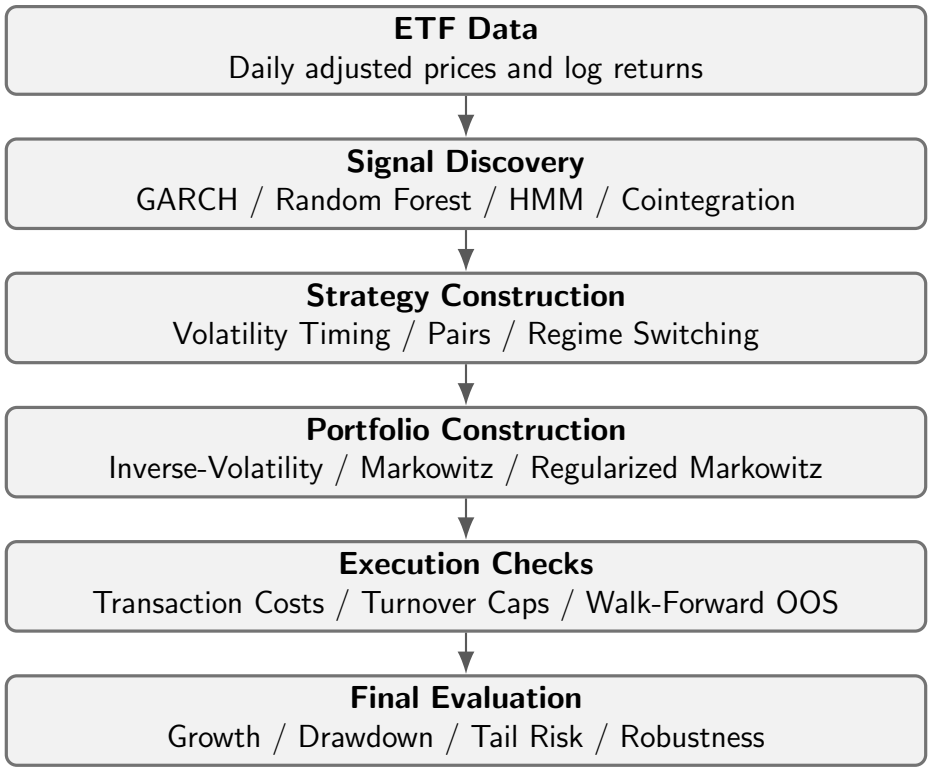


Figure 1: End-to-end project workflow. Each stage narrows the set of ideas before the final portfolio evaluation.

1 Research Question

The central problem is that a signal can pass one layer of analysis and fail at the next. A model can look statistically useful but fail once converted into positions. A strategy can work alone but add little value after portfolio construction. A portfolio optimizer can look strong before costs and become fragile after realistic execution assumptions.

This project is designed around that practical challenge. It evaluates systematic trading ideas through three linked questions:

1. **Does the signal exist?** Test volatility predictability, spread mean reversion, and market regimes.
2. **Can the signal be traded?** Convert the signal into positions and measure after-cost strategy performance.
3. **Does it improve a portfolio?** Combine retained strategies under risk targets, turnover caps, transaction costs, and slippage.

This structure keeps the work honest: a weak signal is allowed to fail, and a strong-looking backtest is not accepted until it survives cost and validation checks.

2 Research Design

The project begins with adjusted close prices and daily log returns. SPY is used as the main benchmark because it is a liquid proxy for broad U.S. equity-market risk. The configured pair is EFA–EEM. The annualization factor is 252, the transaction-cost assumption is 5 basis points, the slippage assumption is 2 basis points, the portfolio turnover cap is 0.35, and the final target portfolio volatility is 10% annualized.

Research block	Question tested	Output used later
Volatility forecasting	Can next-period SPY risk be forecast from lagged volatility and return features?	A volatility-timing rule that scales SPY exposure.
Cointegration and OU spread	Is EFA–EEM a stable mean-reverting pair?	A pairs-trading rule only if the spread passes the research filter.
HMM regimes	Can SPY returns be separated into calm, intermediate, and stress-like regimes?	A regime-switching rule that reduces exposure in defensive states.
Portfolio allocation	Do the retained strategies improve a combined book after costs and trading constraints?	Inverse-volatility and Markowitz allocation variants.

Table 1: Research design: each model is linked to a trading or portfolio decision.

The detailed report gives the formal metric definitions. Here, the metrics are used in a practical way: Sharpe and Sortino summarize risk-adjusted performance, CAGR measures compounded growth, maximum drawdown and Calmar describe path risk, and VaR/Expected Shortfall capture bad-day tail severity.

3 Findings

Volatility Forecasts Were Useful for Risk Management

Both GARCH and random forest models tracked realized volatility. The random forest performed best because it used recent realized-volatility information, lagged returns, and rolling features. The result should be interpreted carefully: it is a risk forecast, not a return-direction forecast. High volatility-forecast correlation is plausible because realized volatility is persistent and adjacent realized-volatility windows overlap, but it does not mean the model predicts whether SPY will rise or fall.

Model	MSE	MAE	Correlation
Random Forest	0.000188	0.008328	0.985481
GARCH$(1,1)$	0.027052	0.136491	0.901712

Table 2: Volatility forecast comparison. Lower MSE and MAE indicate smaller forecast errors.

When converted into a trading rule, volatility timing modestly improved the return path relative to naive SPY buy-and-hold. It kept meaningful upside participation while scaling exposure when forecast risk rose.

The EFA–EEM Pair Failed, and That Was Useful

The EFA–EEM pair failed the cointegration test and then performed poorly when traded. That negative result is useful because the statistical test and the backtest pointed in the same direction. The pair was not forced into the final portfolio simply because it had been researched. It was rejected because the evidence did not support it.

Regime Switching Added Defensive Behavior

The HMM identified calm, intermediate, and crisis-like market states. The regime-switching strategy reduced exposure in defensive regimes, which lowered volatility, drawdown, VaR, and Expected Shortfall. The tradeoff was lower CAGR because the strategy gave up some upside during rebounds and bull markets.

	Vol. Timing		Buy-and-Hold		Pairs Trading		Regime Switching	
Metric	IS	OOS	IS	OOS	IS	OOS	IS	OOS
Sharpe ratio	1.19	0.92	1.16	0.76	0.05	-0.24	1.08	0.83
Sortino ratio	1.70	1.13	1.61	0.89	0.05	-0.27	1.21	0.90
CAGR (%)	20.90	9.99	19.87	13.06	0.04	-2.50	8.45	5.99
Max drawdown (%)	-17.23	-15.26	-19.21	-35.75	-20.32	-28.99	-9.84	-11.60
Calmar ratio	1.21	0.65	1.03	0.37	0.00	-0.09	0.86	0.52
Volatility (%)	17.16	11.04	16.82	18.26	8.58	8.87	7.78	7.35
VaR_{95} (%)	1.73	1.09	1.58	1.71	0.86	0.95	0.72	0.68
ES_{95} (%)	2.44	1.76	2.33	2.85	1.36	1.43	1.25	1.18

Table 3: Single-strategy results with in-sample (IS) and walk-forward out-of-sample (OOS) performance after transaction costs and slippage.

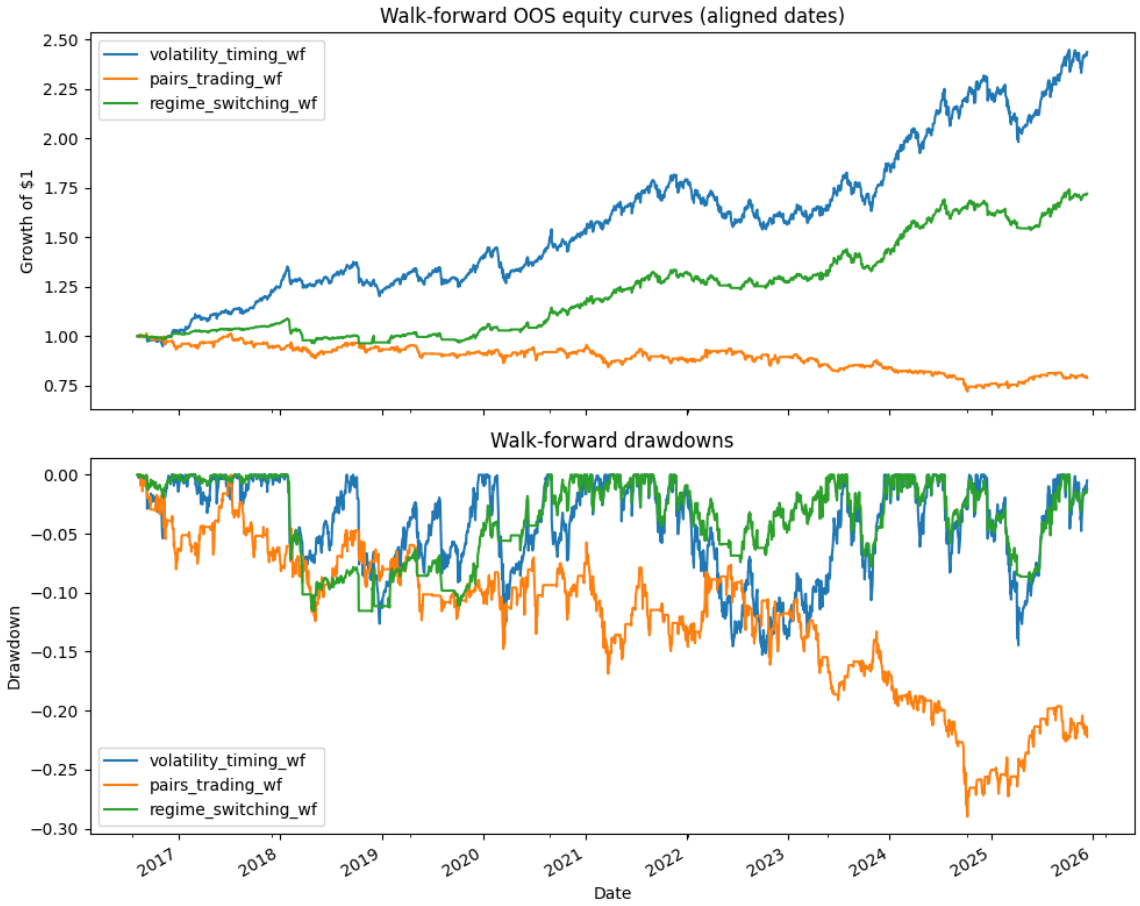


Figure 2: Walk-forward out-of-sample equity and drawdown curves. Volatility timing maintains the best growth profile, regime switching remains more defensive, and pairs trading continues to deteriorate.

4 Portfolio Construction and Execution

The final portfolio uses only the retained strategy sleeves. The failed EFA–EEM pair is excluded, so the allocation step focuses on the volatility-timing and regime-switching return streams. This prevents the final results from relying on a weak strategy just for diversification optics.

Three selected allocation methods are compared:

1. **Inverse-volatility weighting**: allocates more to the lower-volatility sleeve and does not forecast expected returns.
2. **Markowitz baseline**: uses mean–variance optimization but can become sensitive to noisy inputs.
3. **Regularized Markowitz**: adds a concentration penalty to discourage extreme allocations.

The portfolio test includes transaction costs, slippage, turnover caps, walk-forward evaluation, and volatility targeting. These assumptions reduce paper performance, especially for optimizer-driven allocations, but they make the comparison closer to how a strategy would actually be judged.

	Inverse Vol.		Markowitz Base		Markowitz Reg.	
Metric	IS	OOS	IS	OOS	IS	OOS
Sharpe ratio	1.25	0.79	0.99	0.85	1.11	0.84
Sortino ratio	1.51	0.87	1.16	0.99	1.35	0.99
CAGR (%)	13.49	8.23	10.45	8.86	11.80	8.68
Max drawdown (%)	-10.47	-17.51	-11.46	-17.13	-11.30	-13.57
Calmar ratio	1.29	0.47	0.91	0.52	1.04	0.64
Volatility (%)	10.55	10.78	10.67	10.73	10.50	10.65
VaR_{95} (%)	0.94	0.96	1.05	1.00	1.02	0.99
ES_{95} (%)	1.60	1.71	1.70	1.70	1.62	1.68

Table 4: Selected after-cost portfolio comparison with full in-sample and walk-forward out-of-sample metrics.

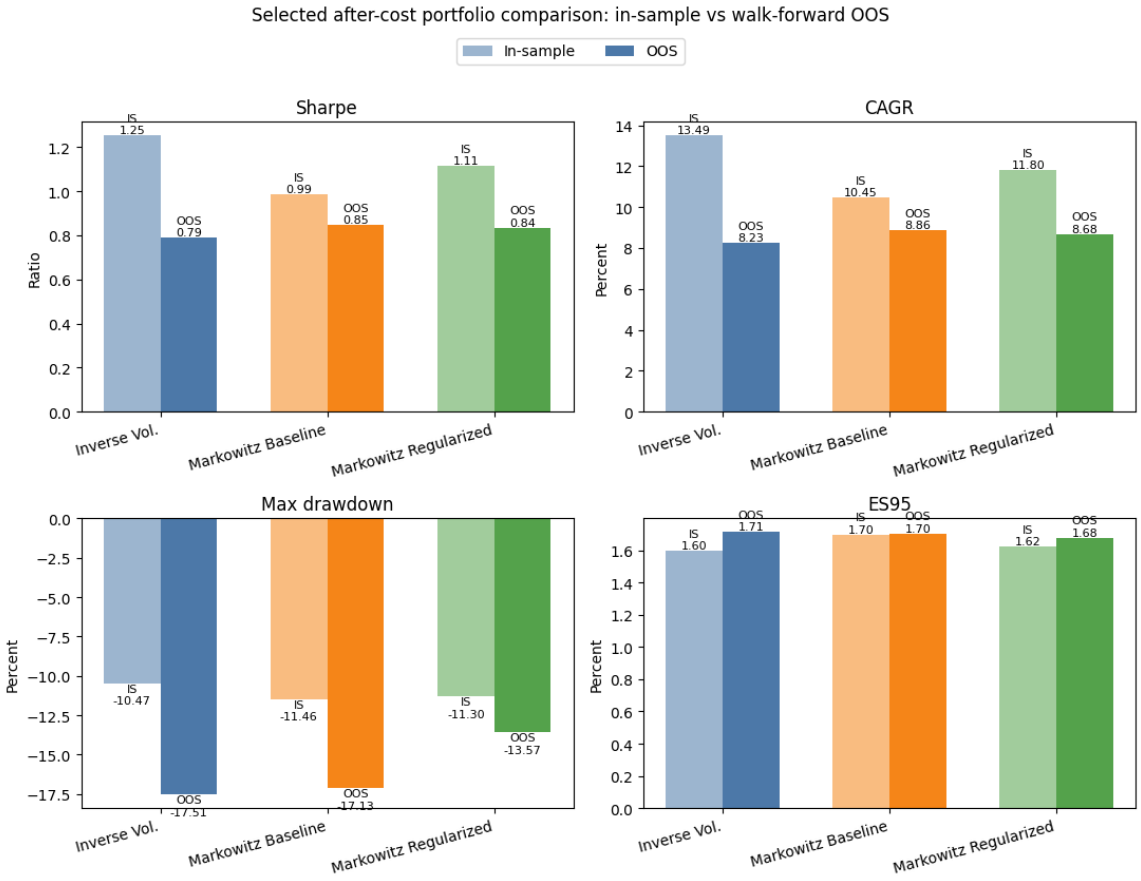


Figure 3: Selected after-cost in-sample portfolio comparison. Inverse-volatility has the strongest overall tradeoff, while concentration regularization improves Markowitz baseline performance but does not fully close the gap.

Inverse-volatility weighting was the most reliable selected allocator after costs because it is simple, slow-moving, and less exposed to noisy expected-return estimates. Markowitz baseline was competitive out of sample, but it suffered large drawdowns and required additional controls. Regularization improved optimizer stability and produced the smallest selected OOS drawdown, even though it did not dominate every return metric.

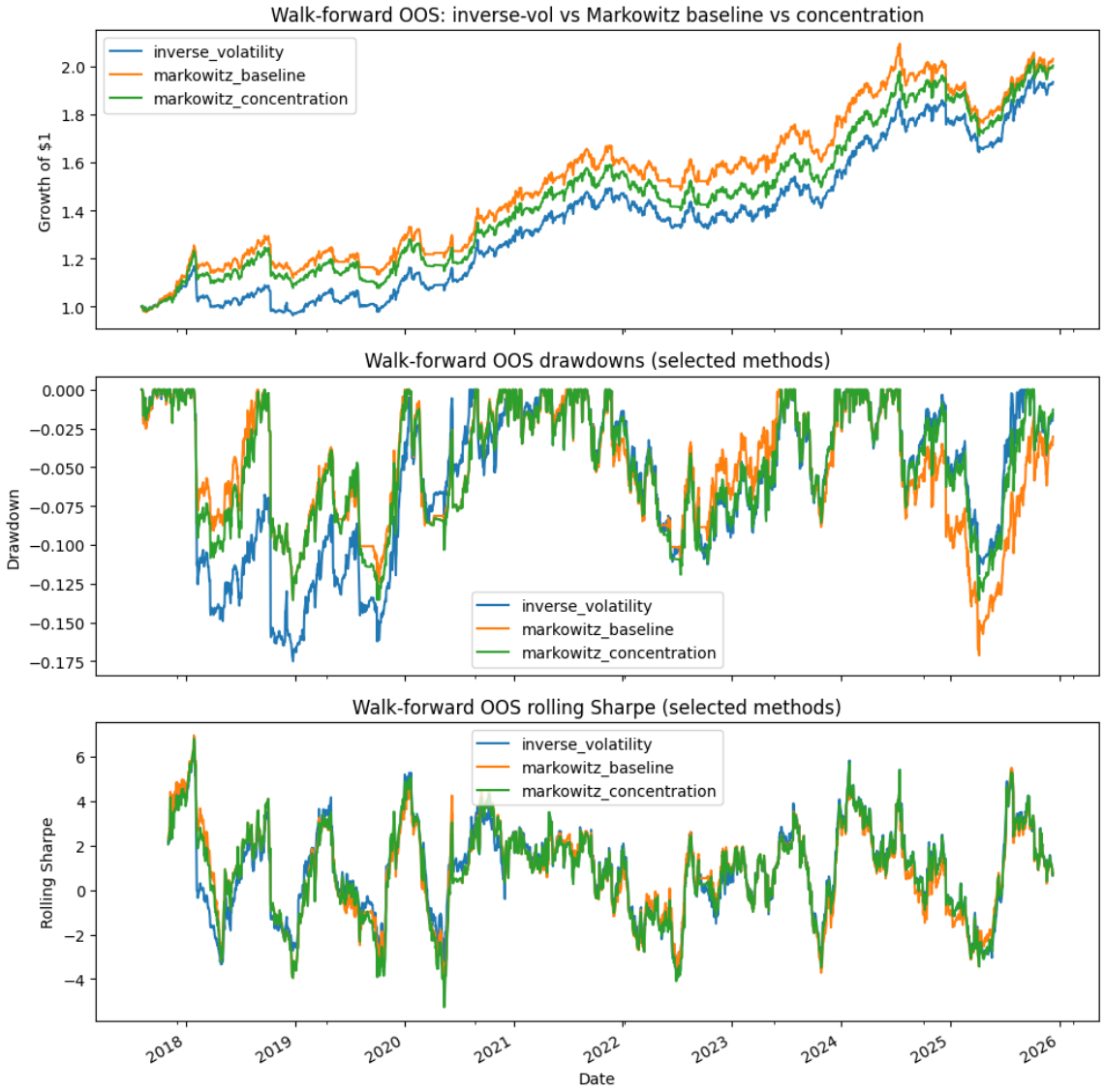


Figure 4: Walk-forward out-of-sample portfolio equity, drawdown, and rolling-Sharpe curves after turnover caps, transaction costs, slippage, and volatility targeting for the three selected portfolio methods.

5 Robustness and Model Credibility

The project includes several safeguards against overly optimistic conclusions. The model tests use chronological splits rather than random shuffling. The random-forest volatility target is shifted one step ahead, and features are formed from information available at the forecast date. The strategy and portfolio layers include walk-forward testing, costs, slippage, turnover caps, and volatility targeting.

These checks do not eliminate all overfitting risk. The ETF universe is compact, the sample

is one historical path, strategy thresholds are intentionally simple, and market regimes may not remain stable in the future. The correct interpretation is therefore measured: volatility and regime information were useful in this research setting, but the exact rules are or should not be treated as permanent without further testing.

6 Takeaway

This project connects data, models, trading rules, and portfolio evaluation in one chain. The main result is not a single Sharpe ratio; it is the workflow: test the signal, trade it only if the evidence survives, combine it only if it improves the portfolio, and evaluate it after costs and constraints.

Useful models are not the same as useful strategies. The project separates statistical evidence from tradable performance. This is why the EFA–EEM pair could be rejected instead of forced into the final result.

Risk forecasts can be valuable without predicting returns. Volatility timing and regime switching worked mainly by changing exposure when risk conditions changed, not by forecasting daily market direction.

Portfolio construction must include trading frictions. Turnover caps, costs, slippage, and volatility targeting changed the ranking and made simple allocation rules more attractive.

Optimization needs guardrails. Markowitz methods can be useful, but they are sensitive to noisy estimates. Regularization and hard caps are not minor technical details; they are practical safeguards for reducing fragile allocations.

Negative results improve credibility. The failed pairs trade is part of the value of the project because it shows that the screening process can reject weak ideas instead of only presenting successful ones.

In practice, many quantitative failures occur not because the model is mathematically invalid, but because trading costs, unstable optimization, and weak robustness checks are ignored. This project implements those checks directly through signal filtering, walk-forward testing, cost and turnover assumptions, volatility targeting, and regularized portfolio construction.